Exercise I: [15 points=MCQ+True/False] Choose the correct answers

MCQ Questions (10 points = 5x2)

1. The design process for identifying the sub-systems making up a system and the framework for sub-system control and communication is:
    a. Low-level design
    b. Architectural design
    c. Software design
    d. Coding & Programming
2. What defines the repository model?
    a. Data sent peer-to-peer
    b. Central shared database
    c. No data sharing
    d. Sub-systems define own formats
3. Capture and record information about the transactions that affect the organization.
    a. Communication Support Systems
    b. Transaction Processing System
    c. Office Support Systems
    d. Decision Support Systems
4. System structuring is the:
    a. Decomposition of the system into several modules.
    b. Reconstruction of the system from its components.
    c. Decomposition of the system into several principal sub-systems.
    d. Collaboration between all stakeholders.
5. _____________ will be an error in DFD:
    a. A process has no input.
    b. A process has no output.
    c. Data flow between two data store.
    d. All above.

True/False Questions (5 points = 5x1)

1. True or False: In the client-server architecture, the server primarily stores and manages data, while clients request and utilize the data from the server.
2. True or False: The Three-tier architecture is more advanced than the two-tier architecture.
3. True or False: Architectural decomposition design aims to create highly interconnected components within a system.
4. True or False: The main objective of architectural design is to define the implementation details of individual system components.
5. True or False: The primary goal of information system design is to develop a system that meets the functional requirements specified by the stakeholders.

Exercise 2: Use Case Analysis [40 points]

Part A: Use case diagram [25 points]

College Management System (CMS) is an application used by most of the universities worldwide to track the activities of instructors, students, and administrator. The system includes registrations into variety of activities such as course registrations, seminar registrations, etc. The activities are organized as following:

1. The administration department has the following activities:
    a. Admin can create a new semester.
    b. Admin can manage a course and a section. The management includes creating/editing/removing a course or a section. Note that a course cannot be removed unless all its sections are removed; a section cannot be created before the creation of the corresponding course.
    c. Admin can manage a student account through creating/editing/removing the account.
2. The instructor in all department shares the below activities:
    a. An instructor can view the list of course they teach and may print out their schedule.
    b. An instructor can obtain the section roster and may download/print out the roster.
    c. An instructor can send an announcement to all the students registered in a specific section.
    d. An instructor can also schedule a seminar to be given to a group of students. The seminar can be dropped or edited.
3. The student's activities are as following:
    a. A student can add / drop a course. Note that a course cannot be dropped if it was not added.
    b. A student can change the section of a registered course.
    c. A student can view all the list of registered courses and may print it out.
    d. A student receives the payment information from the billing system after the completion of the registration.
    e. A student can make an appointment with the instructor. The latter should acknowledge the student's appointment.
    f. A student can enroll into a seminar. The student has to enter their student id and their name.
    g. After attending the seminar, the student can receive an attendance certificate. The student may then print it out.
    h. A student can cancel a seminar if he/she does not want to attend it anymore.
4. All users have the following common activities:
    a. A user can login/logout to/from the system.
    b. A user can edit their profile.
    c. A user can view the course catalog by specifying the department id and the course code.

Give the use-case model diagram, showing the interaction between actors and each of the use-cases in the system, and showing the dependencies among use cases (includes, extends, or depends on). You have to show any inheritance relation, if it exists

Part B: Use-case narratives [15 points]

Create the use-case narrative for the use case “Enroll into seminar” to document the business requirements of the previous question

Solution:

Use-Case Name:	Enroll into a seminar	
Use-Case ID:	MSC-REQ015	
Priority:		
Primary Actor:		
Description:		
Precondition:		
Typical Course of Events:	**Actor Action**	**System Response**
Conclusion:		

Exercise 3: System design [30 points]

Consider a system for a mobile app / web - based Smart City Parking System that allows city authorities and drivers to:

1. **Manage Parking Availability and Reservations**: System monitors parking in real-time, allows drivers to search/reserve.
2. **Facilitate Parking Enforcement and Payments**: System supports mobile payment, violation checks, and ticket management.

To access system features, *users create an account*. *User authentication* ensures security and personalization. Users manage profiles, preferences, and history.

Answer the following questions, based on the given case study:

Q1 - (10 points):

- **Designate/identify at least two principal subsystems/components** within the web-based Smart City Parking System. For each of these components, briefly describe its main responsibilities and how it interacts with other subsystems/components.
- Give the corresponding **system components diagram (level 1).**

Q2- (10 points) Give the high-level architectural design figure (level 2) of the smart city parking system, by showing how various software components interact.

Q3- **Level3 – Detailed Design** (10 points)::

a. Identify at least two datasets that could be used by the Smart City Parking System. Give the details by listing the possible datasets (5 points):

Dataset Name	Short Description	Solicited/Used by Component(s)

b. Frontend modeling (design) – Design the user interface (GUI) and list the steps for the process of a driver confirming and paying (include a payment option of your choice) for reserving a parking spot using the mobile app. (5 points)

Exercise 4: DFD for driver parking reservation [15 points]

Give the DFD that describes the driver parking reservation process as follows:

- Driver Registration Process:
  - The driver provides their personal and vehicle information.
  - The system validates the information and stores it in the User DS.
  - The system provides a confirmation to the driver.
- Parking Space Search Process:
  - The driver enters their desired location & time.
  - The system retrieves available parking spaces from the Parking Space DS.
  - The system displays the matching parking options to the driver.
- Reserve Parking Process:
  - The driver selects a parking spot.
  - The system checks the availability of the selected spot in the Parking Space DS.
  - If available, the system creates a reservation record in the Reservation DS.
  - The system confirms the reservation to the driver.